

Web application development

Internet and Web



KHOA CÔNG NGHỆ THÔNG TIN
TRƯỜNG ĐẠI HỌC KHOA HỌC TỰ NHIÊN



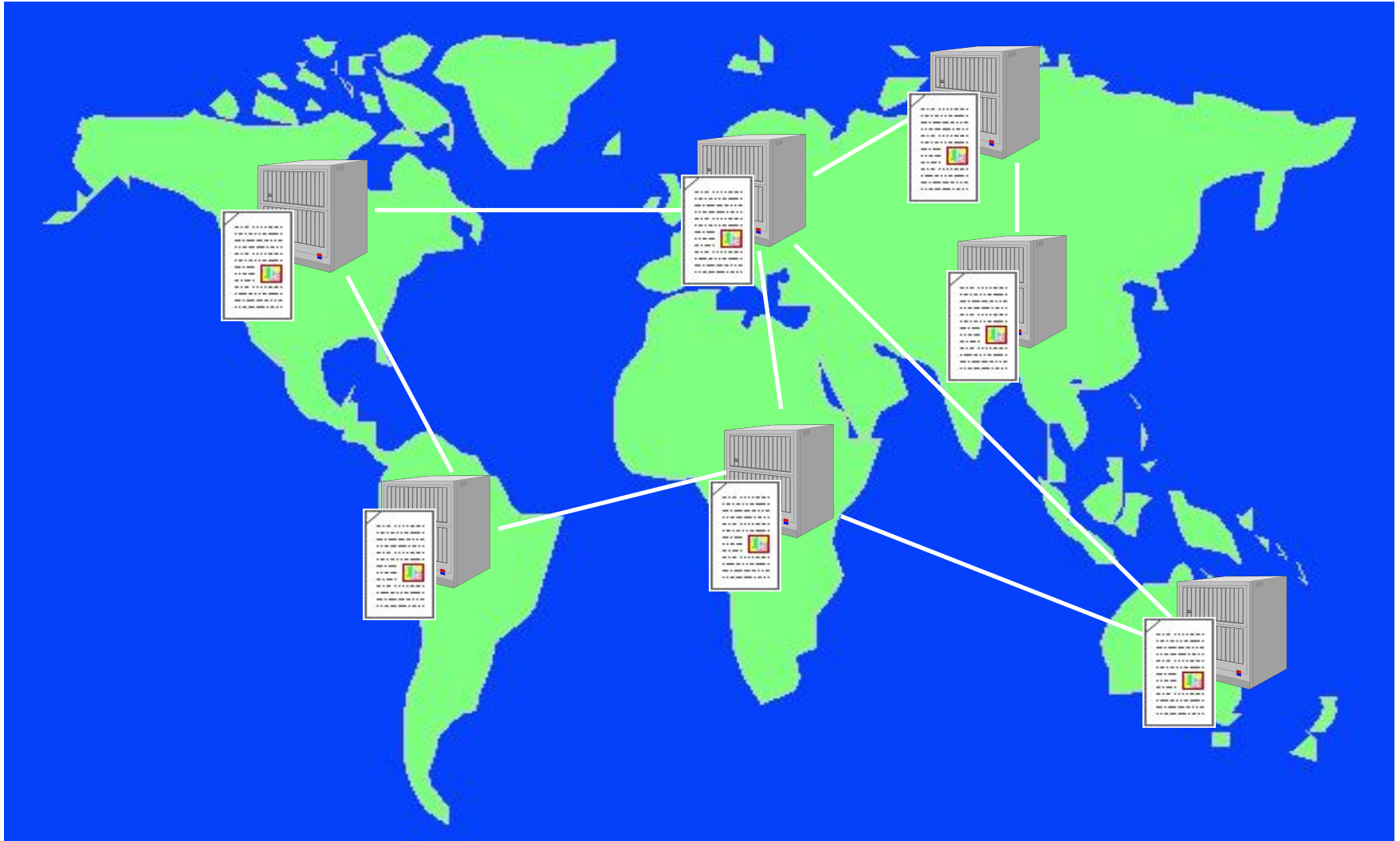
Internet history

Basic concepts

Basic Internet services

Resource Exploration on the Internet

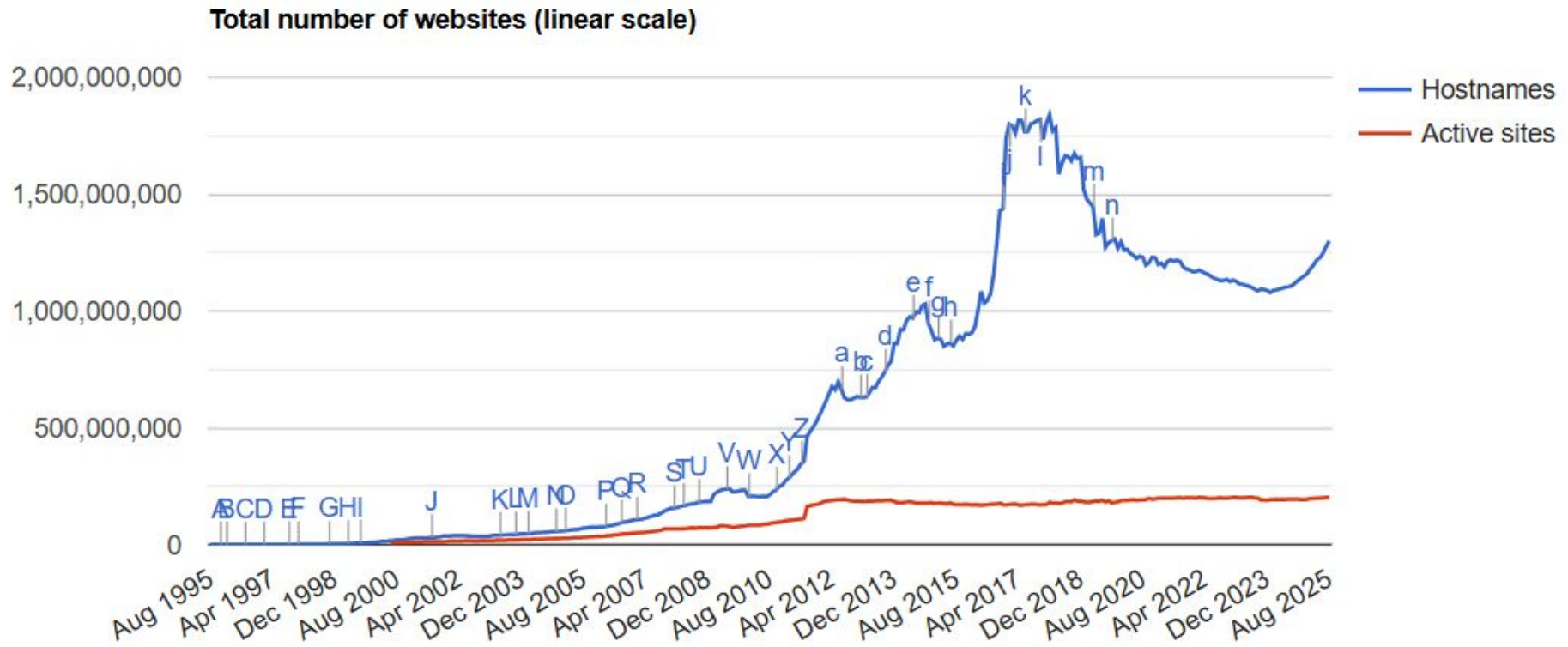
Steps to Set Up a Website





- ☐ The Internet originated from the ARPANET network in the U.S. (1969) – a network connecting various universities.
- ☐ In 1989, Tim Berners-Lee invented the World Wide Web protocol and later created the Hypertext Markup Language (HTML).
- ☐ On December 25, 1990, Tim wrote the first web browser and web server (CERN HTTPd).
- ☐ Since 1993, the Internet has developed rapidly.
- ☐ Vietnam had Internet access in November 1997.

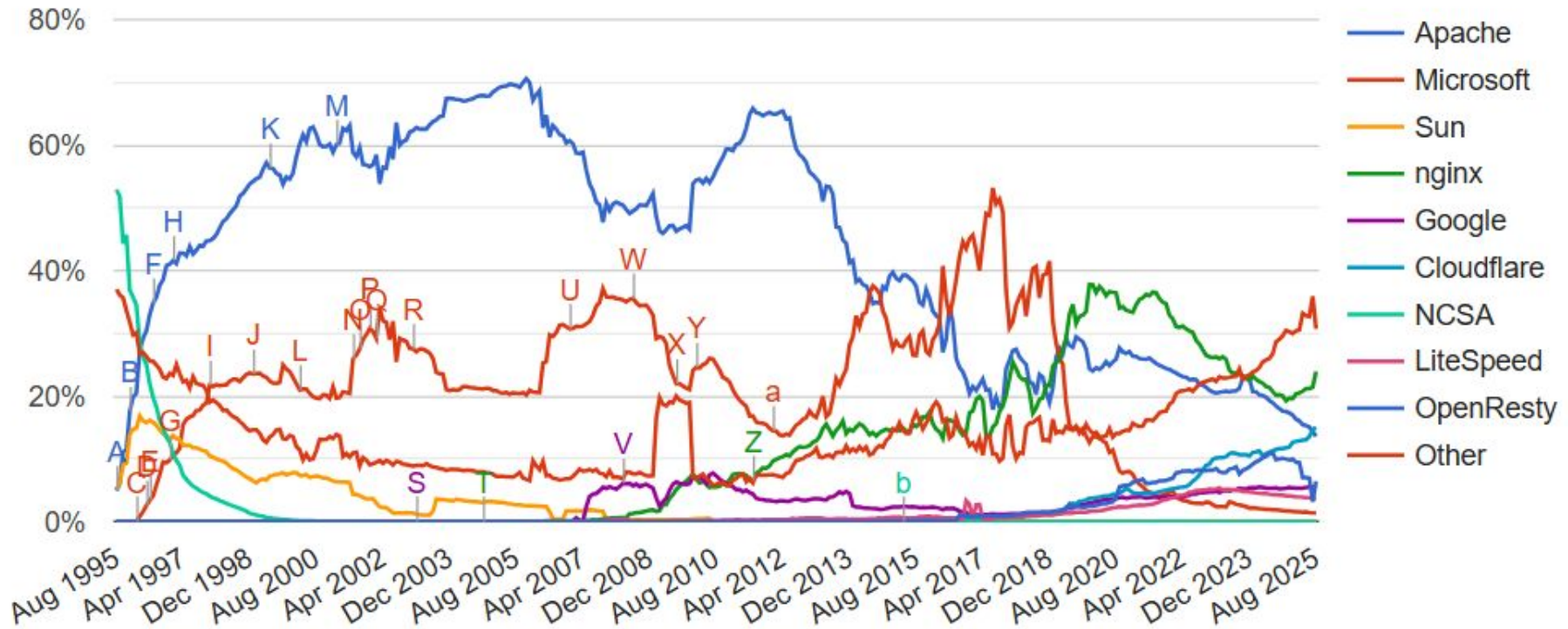
Total number of websites



<https://www.netcraft.com/blog/august-2025-web-server-survey>



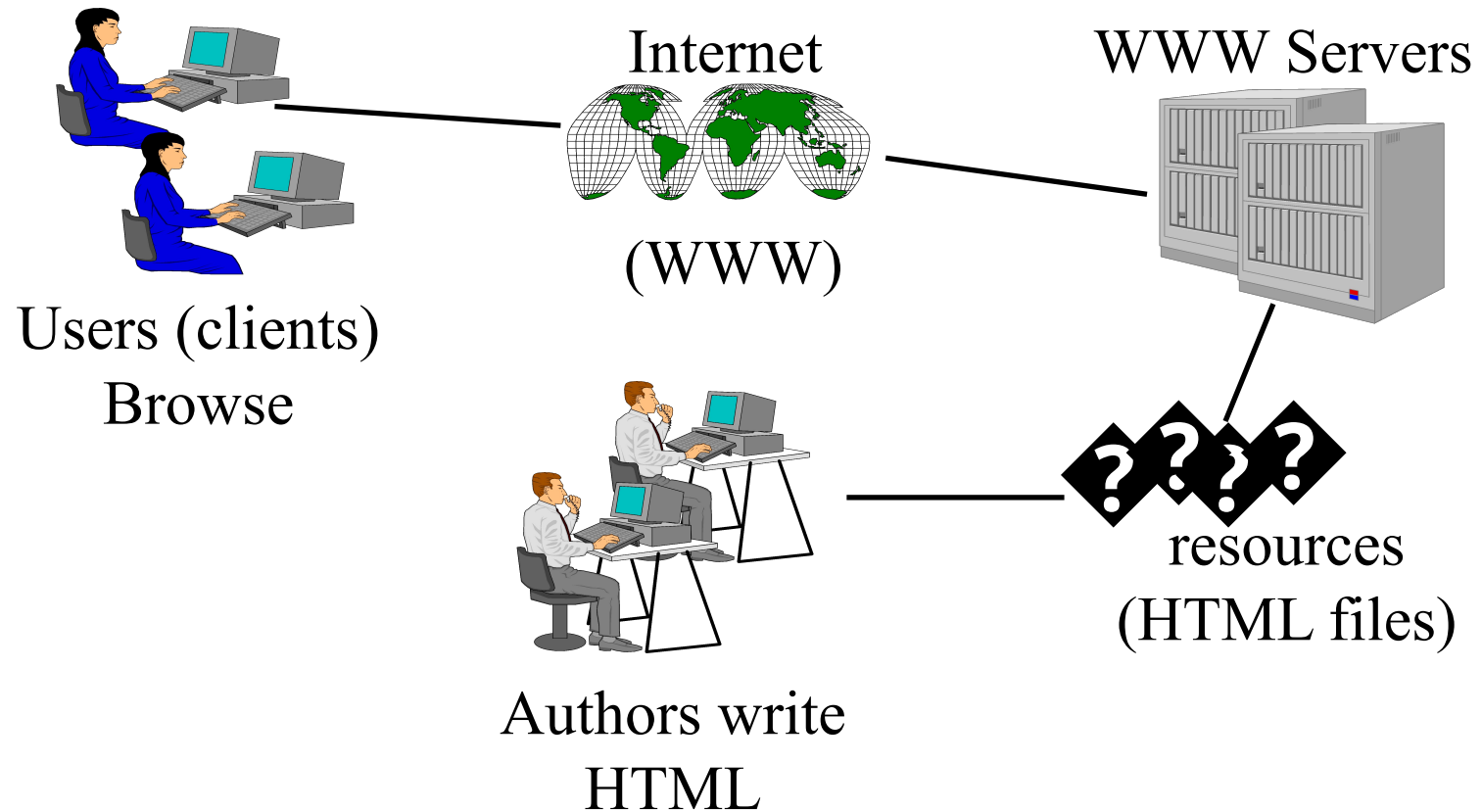
Web server developers: Market share of all sites



<https://www.netcraft.com/blog/august-2025-web-server-survey>



-  These are distributed services that provide multimedia information based on hypertext.
 - ☐ **Distributed:** Information is stored on multiple servers around the world.
 - ☐ **Multimedia:** Information includes text, graphics, sound, and video.
 - ☐ **Hypertext:** A technique used to access information.
-  It provides access to network resources like FTP, News, etc.





☐ **Web** = protocol + language + naming infrastructure

☐ **HTTP - HyperText Transport Protocol**

- ☐ The communication protocol between WWW client and server

☐ **HTML - HyperText Markup Language**

- ☐ The language for WWW documents

☐ **URL - Uniform Resource Locator**

- ☐ Unique web address



IP Address

- ☐ 32 bit, separated in to four 8 bit number, example: 203.162.33.44
- ☐ Identify the recipient to send information on the Internet.
- ☐ How to get the IP address: **ping** www.intel.com

```
C:\WINDOWS\system32\cmd.exe

C:\>ping www.intel.com

Pinging a961.g.akamai.net [125.56.199.26] with 32 bytes of data:

Reply from 125.56.199.26: bytes=32 time=110 ms TTL=64
Reply from 125.56.199.26: bytes=32 time=110 ms TTL=64
Reply from 125.56.199.26: bytes=32 time=110 ms TTL=64
Reply from 125.56.199.26: bytes=32 time=110 ms TTL=64

Ping statistics for 125.56.199.26:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 110ms, Maximum = 113ms, Average = 110ms
```

```
C:\WINDOWS\system32\cmd.exe

C:\>tracert www.intel.com

Tracing route to a961.g.akamai.net [125.56.199.26]
over a maximum of 30 hops:
  0  <1 ms    <1 ms    <1 ms    192.168.1.1
  1  15 ms     14 ms     14 ms     adsl.viettel.vn [115.74.64.1]
  2  11 ms     14 ms     14 ms     203.113.190.1
  3  11 ms     14 ms     15 ms     203.113.158.28
  4  14 ms     14 ms     14 ms     203.113.158.138
  5  77 ms     79 ms     80 ms     pos13-3.cr02.hkg04.pccwbtn.net [63.203.113.158]
  6  75 ms     78 ms     75 ms     TenGE9-2.br01.hkg04.pccwbtn.net [63.203.113.158]
  7  113 ms    110 ms    107 ms    singtel.ge5-13.br01.hkg04.pccwbtn.net [63.203.113.158]
  8  112 ms    110 ms    214 ms    203.208.172.181
  9  111 ms    109 ms    109 ms    ge-4-1-0-0.sngc3-dr1.ix.singtel.com [203.208.172.181]
 10  110 ms    110 ms    113 ms    125.56.199.26
Trace complete.
```



Domain name

- ☐ It is the trade name of a company or organization on the Internet.
- ☐ For example: www.intel.com
 - It is the trade name of a company or organization on the Internet.
 - INTEL, which has an IP address of 125.56.199.27
 - It is a top-level domain name (.com, .org, .edu, .biz, .net).

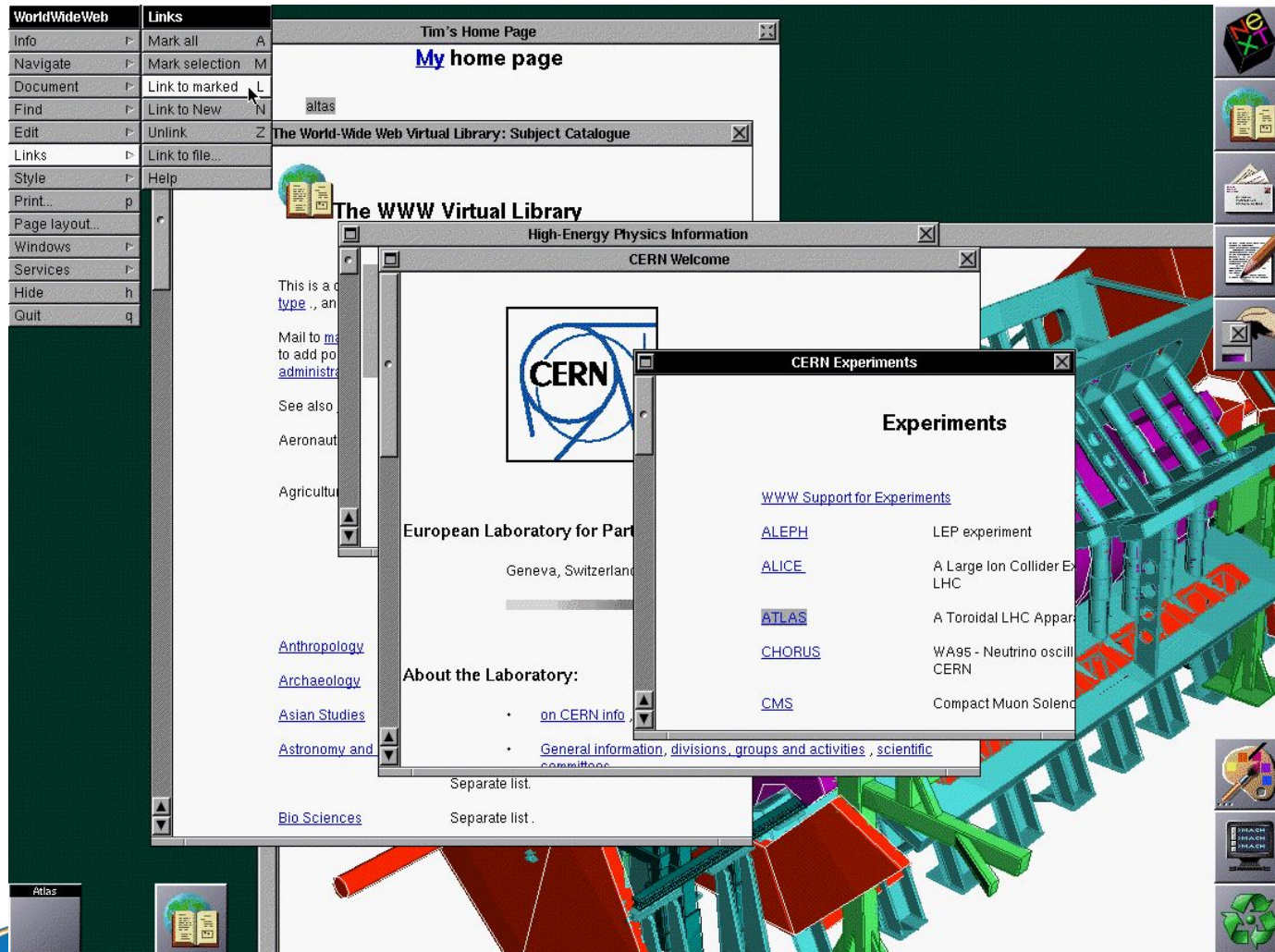


The mapping between the domain name and the IP address, managed by the DNS server – Domain Name System.



- A software application that allows users to access, retrieve, and view content on the World Wide Web.
- Interprets HTML, CSS, and JavaScript to display web pages and provides functionalities such as navigation, bookmarking, and downloading files.
- Popular web browsers include Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

WorldWideWeb (Nexus) - First web browser





```

C:\ http://images.google.com.vn/imghp?hl=en&tab=wi
Preferences <p1 of 5>

Web Images News Groups Scholar Gmail more v
Blogs Calendar Photos Documents Reader Sites
even more ↵

Sign in

Google
Preferences Preferences Help : About Google
Save your preferences when finished and return to search.
Save Preferences

Global Preferences <changes apply to all Google services>

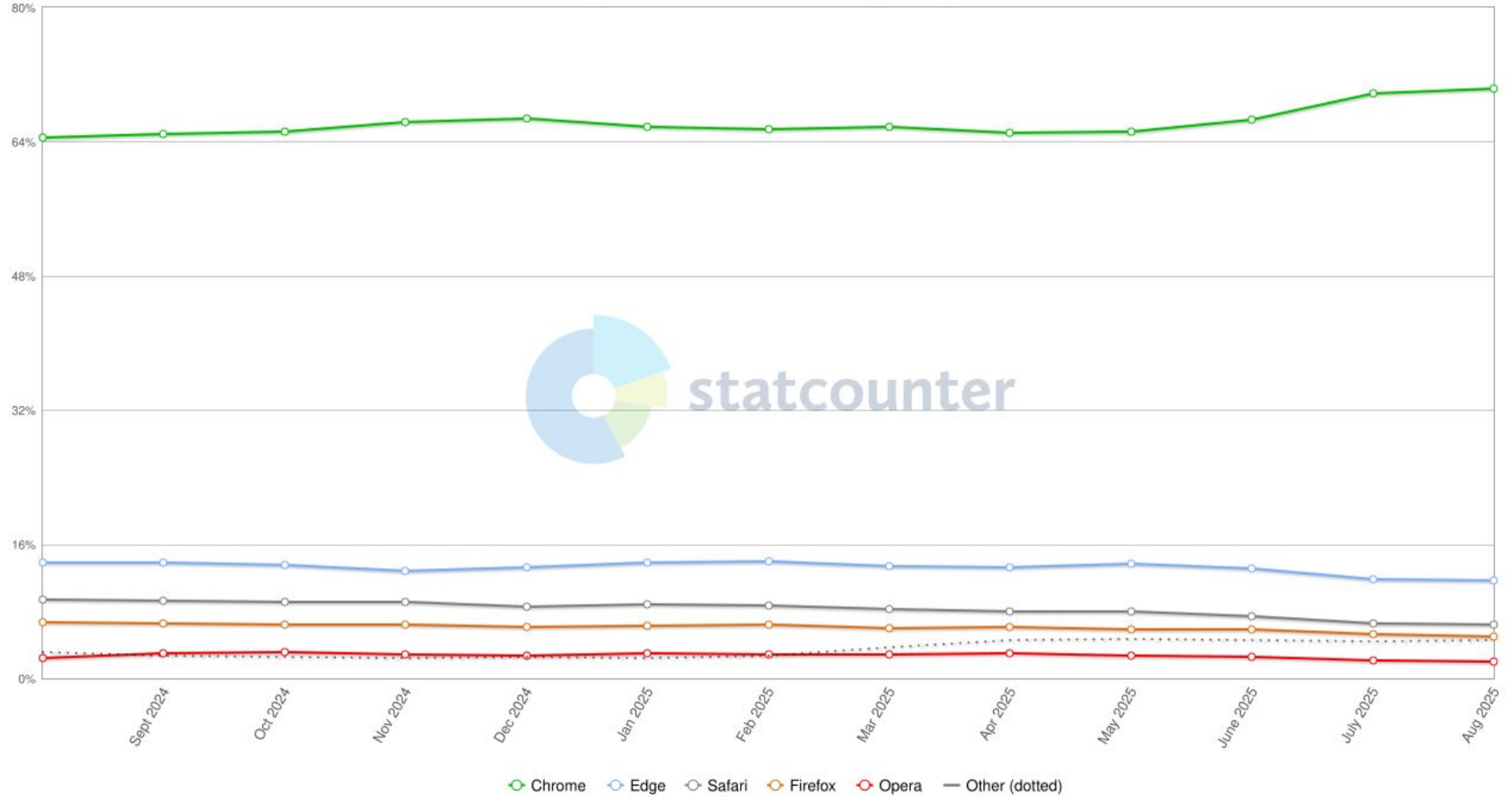
[warning]

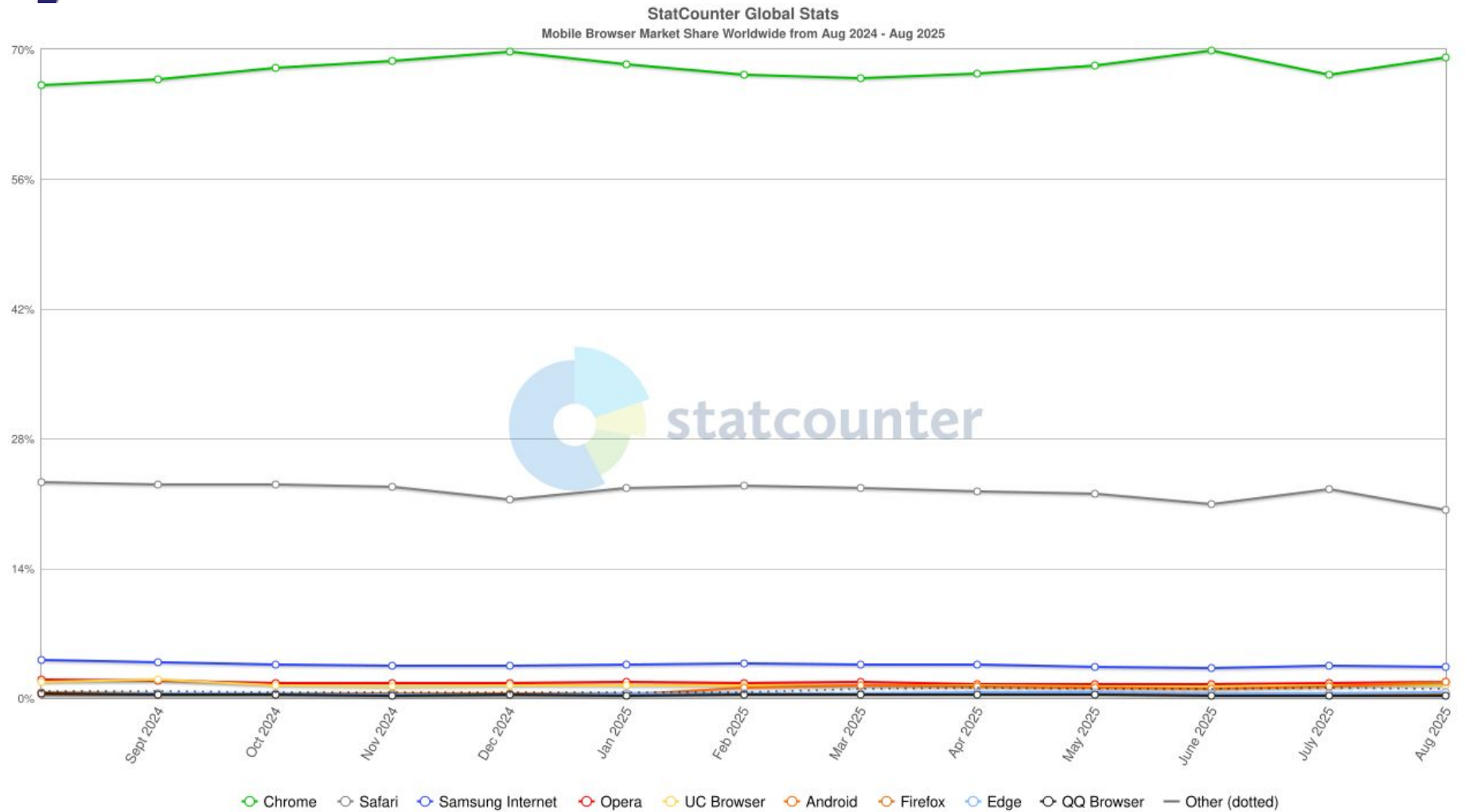
Your cookies seem to be disabled.
Setting preferences will not work until you enable cookies in your
browser.
How do I enable cookies?

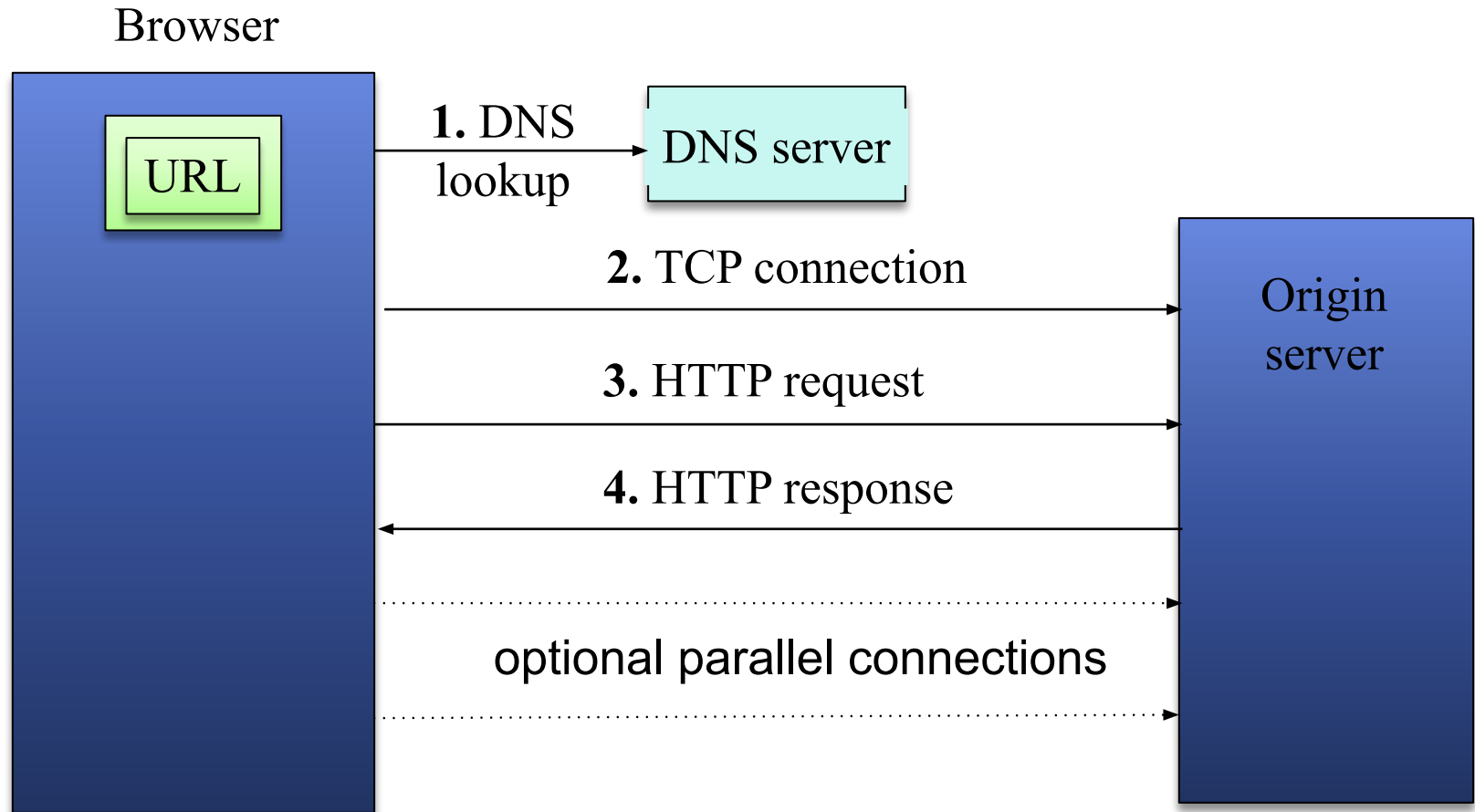
<NORMAL LINK> Use right-arrow or <return> to activate.
Arrow keys: Up and Down to move. Right to follow a link; Left to go back.
H)elp O)ptions P)rint G)o M)ain screen Q)uit /=search [delete]=history list
  
```




StatCounter Global Stats
Desktop Browser Market Share Worldwide from Aug 2024 - Aug 2025









- **Web 1.0** is the earliest stage of the World Wide Web, characterized by static web pages and limited interactivity. Users primarily consumed content without much ability to contribute or interact. It was mainly focused on **information dissemination**, with websites resembling online brochures.
- **Web 2.0** refers to the transition to a more interactive and **user-generated** web. It introduced **social media platforms**, blogs, and collaborative content creation. Users can now create, share, and engage with content, leading to the rise of communities and participatory applications like Facebook, YouTube, and Wikipedia.
- **Web 3.0** is often referred to as the "**semantic web**," where the focus is on creating a more intelligent and interconnected web experience. It aims to utilize **artificial intelligence, machine learning, and blockchain technology** to enhance data interoperability, privacy, and user control. The goal is to provide personalized content and services while enabling greater transparency and decentralization.

Steps to Set Up a Website



- ☐ Determine website requirements
- ☐ Purchase a domain name
- ☐ Develop the website
- ☐ Rent hosting space
- ☐ Launch the website
- ☐ Maintain information and perform website maintenance



- ☐ Survey free hosting services that support Node.js.
- ☐ Register an account.
- ☐ Fill in the host information in the report.
- ☐ Scoring: 3.5 points for each successfully registered host.